



HACKATHON PROJECT OVERVIEW

Project Name: **Translatio**

Problem Statement & ID: AI-Powered Translation Studio
ID: 2

Team Name: **Team** Bharat **AI**

College: VIIT & COEP

City: PUNE



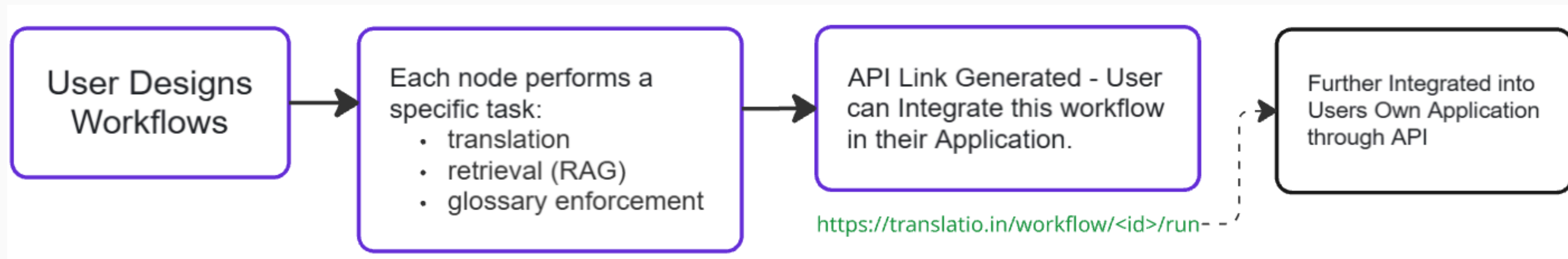
PROPOSED SOLUTION



Translatio is an intelligent platform for building and deploying **Enterprise level translation workflows, tailored for Healthcare Needs**. Think of it as n8n or Zapier, but specifically for creating translation workflows.

Instead of a fixed translation tool, our system allows organizations to build **custom pipelines combining validation, translation, glossary enforcement, and learning**.

How the Solution Works



Key Components:

- Workflow generator UI + Workflow Engine (graph-based execution)
- Node System (modular AI tasks) - Separate Node for each Unique Task
- Translation Memory (RAG) + Feedback Learning System
- Glossary and Style Engine



APPROACH

Step-by-Step Approach

- 1. Define Workflow:** Users design a translation pipeline using modular nodes (validation, RAG retrieval, translation, glossary enforcement).
- 2. Document Ingestion:** Upload and parse documents (PDF/DOCX) into structured segments.
- 3. Workflow Execution :** The backend engine executes nodes sequentially according to the workflow graph.
- 4. Processing Pipeline**
 - Source validation (spelling, formatting, terminology checks)
 - Translation memory retrieval using RAG
 - Glossary and style rule enforcement
- 5. Review and Approval :** Translations are presented in a side-by-side editor for human review.
- 6. Learning and Improvement :** Approved corrections update the translation memory and glossary, improving future translations.

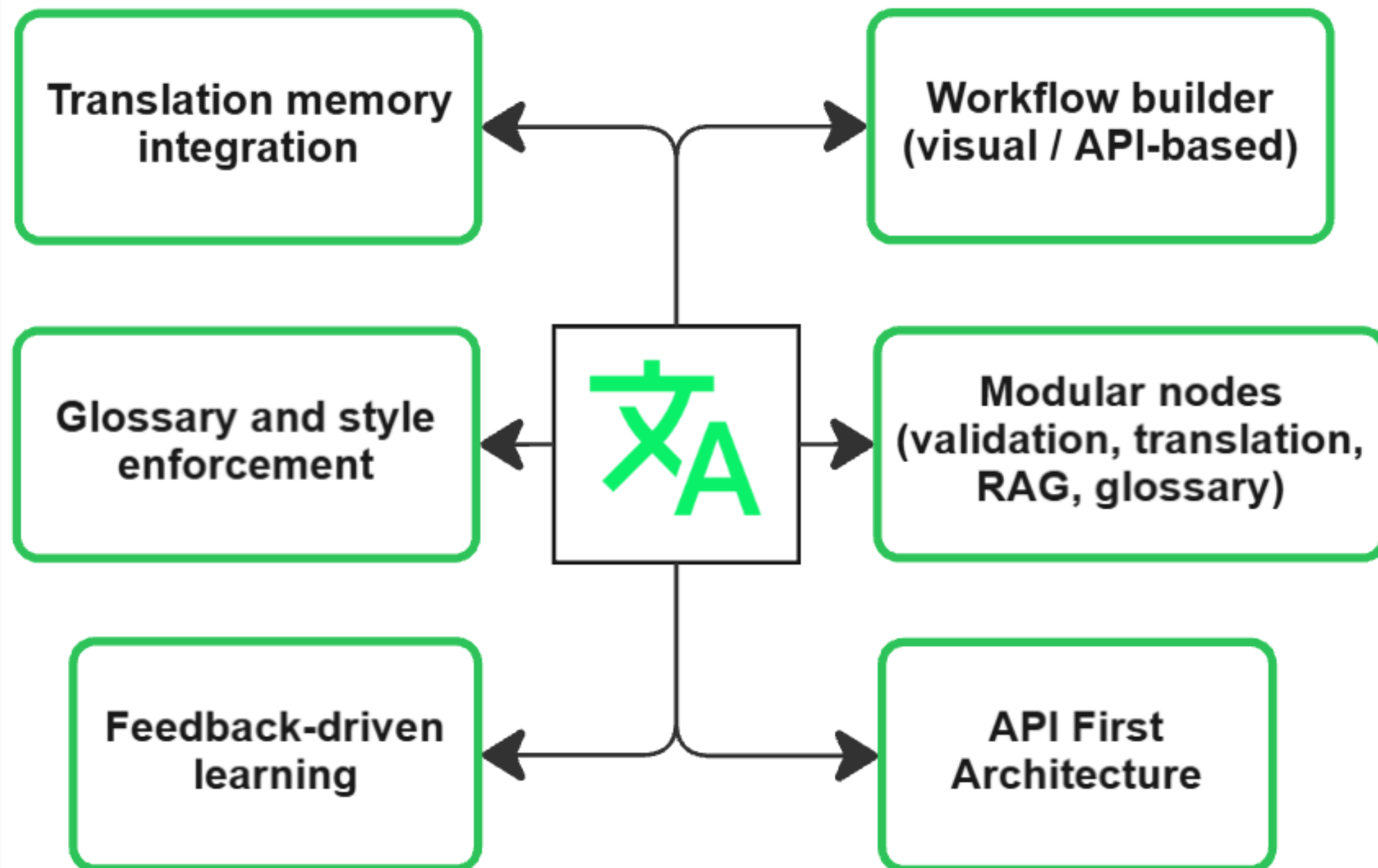
Challenges & Solutions

Challenge	Solution
Rigid translation tools	Custom workflow builder
Inconsistent terminology	Glossary and validation nodes
Repeated translation work	RAG-based translation memory
Lack of flexibility	Modular node-based architecture
No continuous improvement	Feedback-driven learning loop

USP

- Our platform's key distinction is its **workflow-based architecture**, offering far more flexibility than traditional translation tools.
- Instead of a fixed solution, enterprises can **design custom translation pipelines** using modular nodes such as validation, RAG-based translation memory, glossary enforcement, and human review.
- This **building-block approach** solves core translation challenges by improving source quality, **reusing past translations**, **ensuring terminology consistency**, and **enabling continuous learning**.
- As a result, organizations can **deploy scalable, customizable translation workflows** tailored to their specific processes.

Key Features



TECHNOLOGIES USED AND IMPLEMENTATION



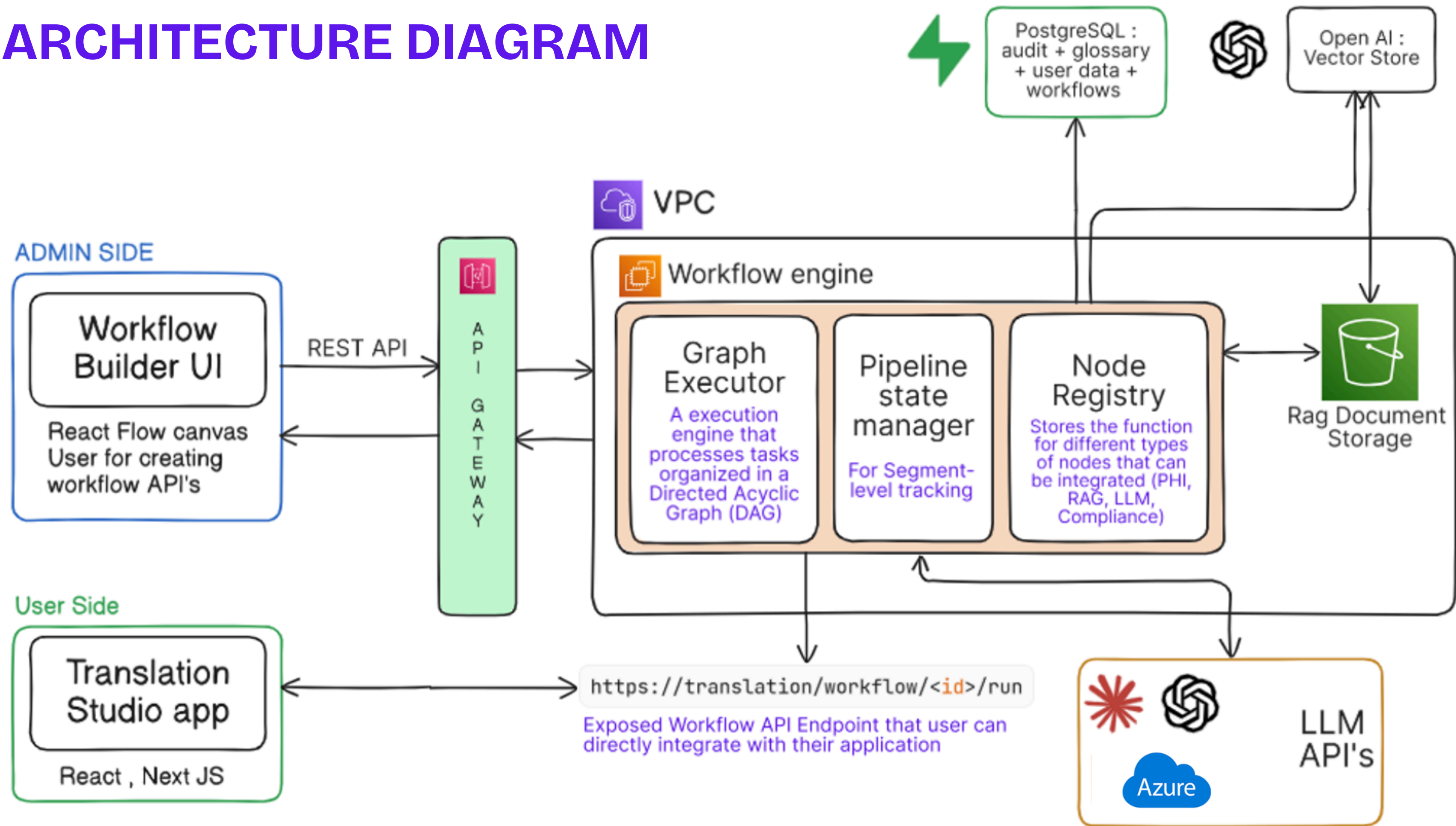
How It Works :

- **Ingest & protect** — Document is parsed into segments. PHI (names, MRNs, DOBs) is masked before any model sees the text.
- **Retrieve or translate** — Approved translations are fetched from vector memory (pgvector + RAG) instantly. Only genuinely new segments reach the ML/LLM layer — IndicTrans2/ **Sarvam** for Indic languages locally at ~80ms, **Claude/OpenAI** for all other pairs with glossary enforced.
- **Score & comply** — COMETKiwi scores every output in ~50ms without a reference. Low-confidence segments are retried. A rule engine **checks CMS and FDA compliance** before any segment reaches a human.
- **Review & learn** — Linguist approves in the side-by-side editor. **Every approval updates the translation memory. The next run is faster and cheaper.**

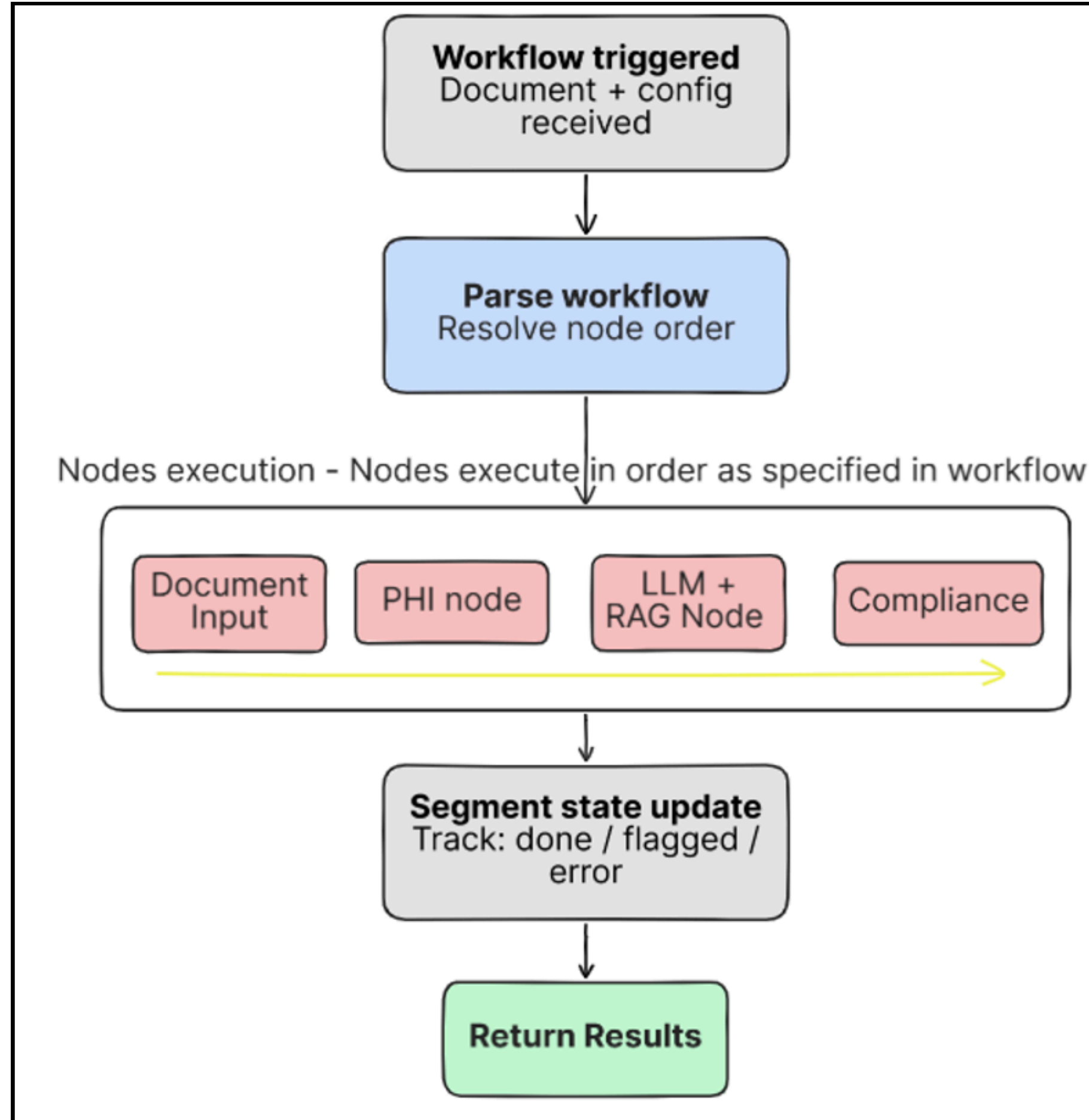
Key Components of Solution :

- **Workflow engine** — DAG-based node executor, every workflow becomes a REST API endpoint.
- **5 core nodes** — PHI Detector , Source Validator , RAG/TM , ML+LLM Translation , Compliance Checker.
- **Translation memory(TM)** — pgvector + multilingual-e5 embeddings, sub-20ms retrieval.
- **ML layer** — COMETKiwi (quality gate) + IndicTrans2 (Indic languages, local, free). Translation Studio app — built entirely with our own APIs. Proof the platform works.

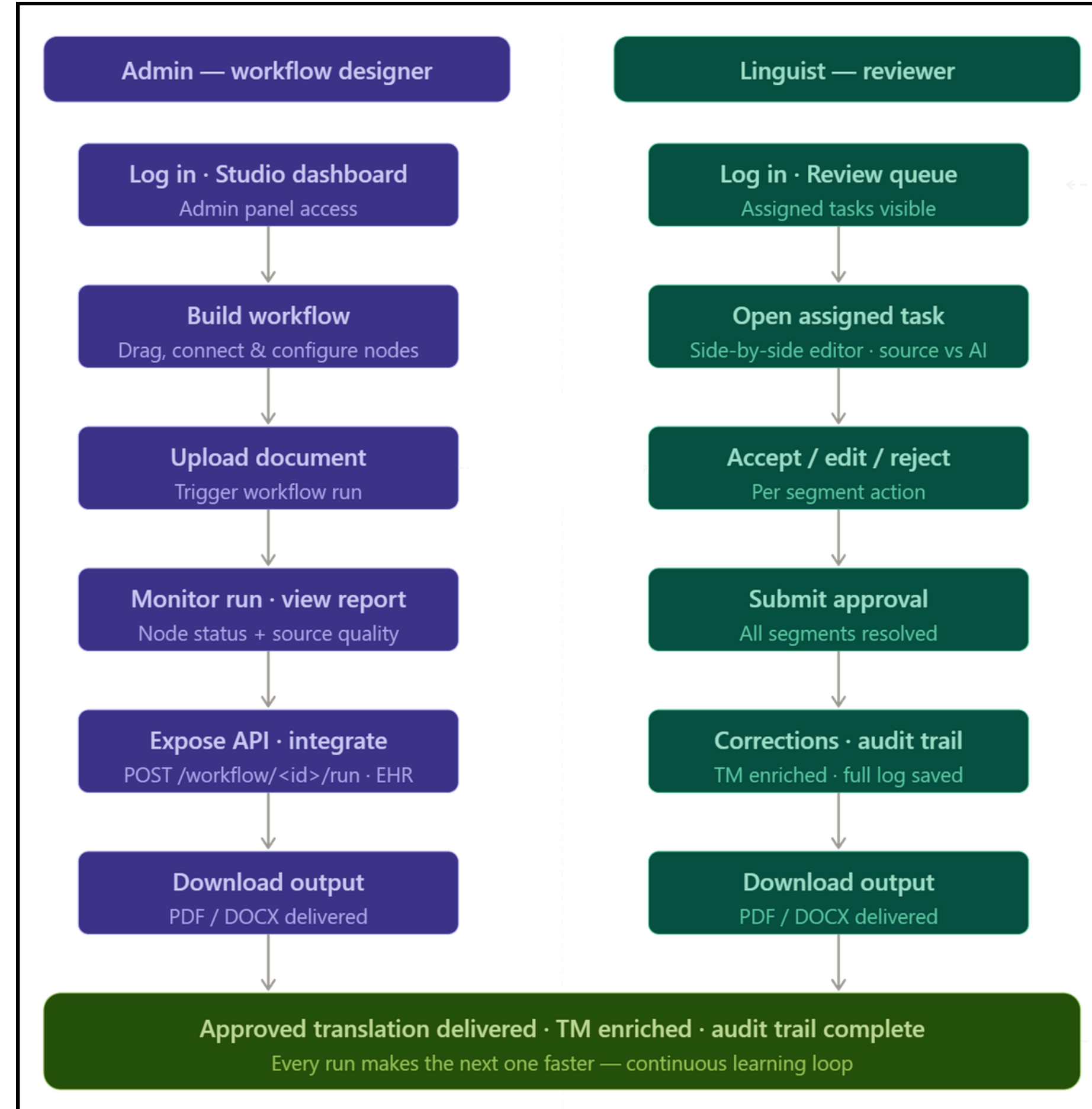
ARCHITECTURE DIAGRAM



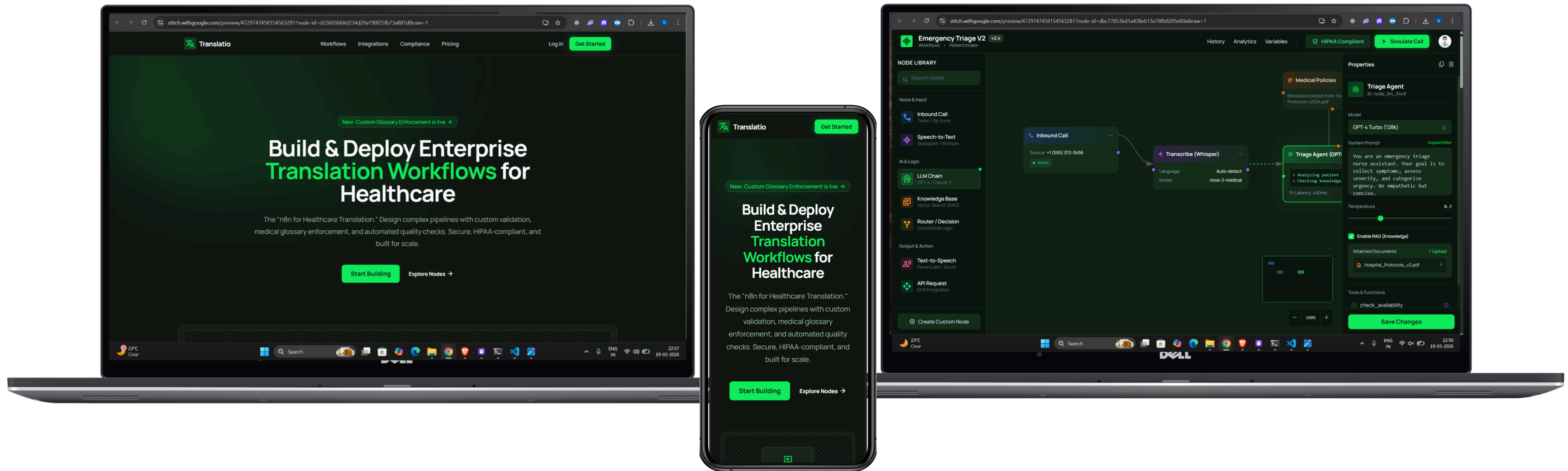
Workflow Execution Flow



USERFLOW DIAGRAM



Project ScreenShots & performance report (estimates)



1. Latency

- TM exact matches at ~40ms, IndicTrans2 locally at ~120ms — no API round-trip for Indic languages. 40–50% of segments never hit the LLM, making repeat document types feel near-instant.

2. Accuracy & Reliability:

- RAG pulls your organization's own approved translations as context at inference time — not generic training data. COMETKiwi catches low-quality output before it reaches a human. TM(Translation Memory) accuracy compounds with every approval and improves translation quality.

3. Scalability & Cost:

- TM hits, IndicTrans2, and COMETKiwi are all free — LLM calls only happen when necessary.
- Every workflow is a REST API endpoint — easy to integrate and endlessly customizable.

TECHNOLOGIES USED

Tech Stack

Development :
React Native with Expo Router
React Flow: For Visualization
Nativewind: For Styling



Cloud Hosting:
Docker: For Containerization
AWS: For Various Hosting Services



Backend & Database :
SupaBase : For Storing User Data
FastAPI : For Efficient Backend
Langchain: For Rag based Chatbot



AI Models : GPT-5, Sonnet 4.6, sarvam-30b
(for Indian Language Translations)
ML Models: COMETKiwi, IndicTrans2



POTENTIAL IMPACT



Impact on Target Audience

1. Hospitals & clinics save 40–60% on translation costs as TM(Translation Memory) leverage grows — smaller hospitals that couldn't afford localization teams now can
2. Linguists spend time on nuance, not repetition — **TM(Translation Memory) auto-fills previous occurred segments**, COMET flags errors before review, human effort focuses where it matters
3. **Compliance officers get a timestamped, immutable audit trail per segment** — any regulatory query answered in seconds, not days

Benefits of the Solution

1. Faster turnaround — 60–70% of segments in repeat document types never hit an LLM, making discharge summaries and consent forms translatable in minutes not hours
2. Higher consistency — **same term is always translated the same way across every document**, every linguist, every language pair
3. Plug-in ready — every **workflow exposes a REST API**, so hospitals integrate Translatio into Epic, Cerner, or any CMS portal without replacing existing systems

Social Impact (India Specific):

1. **India-specific** — Brings **clinical-quality translation** to Hindi, Marathi and Telugu speakers, addressing a gap that affects hundreds of millions of patients who receive healthcare documentation only in English

REFERENCES & LINKS



Research Papers

- 1.COMETKiwi — Reference-free Translation Quality Estimation: arxiv.org/abs/2209.06243
- 2.XCOMET — Explainable Error Span Detection: arxiv.org/abs/2310.10482
- 3.IndicTrans2 — Indic Language Translation: arxiv.org/abs/2305.16307
- 4.Quality-Aware Translation Models (validates score-first approach): arxiv.org/abs/2310.06707
- 5.Calibrating Translation Decoding with QE (2025): arxiv.org/abs/2504.19044

Model Links

- 1.Unbabel/wmt22-cometkiwi-da — huggingface.co/Unbabel/wmt22-cometkiwi-da
- 2.Unbabel/XCOMET-XL — huggingface.co/Unbabel/XCOMET-XL
- 3.Helsinki-NLP/opus-mt-en-hi — huggingface.co/Helsinki-NLP/opus-mt-en-hi
- 4.ai4bharat/indictrans2-en-indic-dist-200M — huggingface.co/ai4bharat/indictrans2-en-indic-dist-200M

Reference Websites

- 1.Anthropic Claude API — docs.anthropic.com
- 2.Sarvam AI (Indic LLM) — sarvam.ai
- 3.pgvector (vector similarity search) — github.com/pgvector/pgvector
- 4.n8n (workflow automation reference) — n8n.io
- 5.Zapier (workflow automation reference) — zapier.com

Team Details



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